I. Particle Swarm Optimization

Pseudo-code of Particle Swarm Optimization (PSO)

Inputs

$\boldsymbol{w}, \boldsymbol{c_1}, \boldsymbol{c_2}$: constants, $\boldsymbol{T}$: max iteration, $\boldsymbol{f}$: fitness function, $\boldsymbol{x_{min}}, \boldsymbol{x_{max}}$: lower and upper bounds

Generate a random population $\boldsymbol{P}$ of $\boldsymbol{N}$ solutions $\boldsymbol{x^{(1)}}, \boldsymbol{x^{(2)}}, \dots, \boldsymbol{x^{(N)}}$ where $\boldsymbol{x^{(i)}} = \left(\boldsymbol{x_1^{(i)}},\ \boldsymbol{x_2^{(i)}}, \dots, \boldsymbol{x_D^{(i)}}\right)$

Initialize the velocity matrix $\boldsymbol{V} \in \mathbb{R}^{\boldsymbol{N \times D}}$ to zero, where each row $\boldsymbol{v^{(i)}}$ corresponds to the velocity vector associated with particle (solution) $\boldsymbol{x^{(i)}}$.

Evaluate the population $\boldsymbol{P}$ and select the best solution $\boldsymbol{x^*}$ as follows :

For $\boldsymbol{i \leftarrow 1}$ **to** $\boldsymbol{N}$ **do**

  IF $\boldsymbol{i = 1}$ **THEN** $\boldsymbol{x^* \leftarrow x^{(i)}}$

    ELSE IF $\boldsymbol{f(x^*) > f\left(x^{(i)}\right)}$ **THEN** $\boldsymbol{x^* \leftarrow x^{(i)}}$ **END IF;**

  END IF;

  $\boldsymbol{x^{(i,p)} = x^{(i)}}$ # Save in $\boldsymbol{x^{(i,p)}}$ personal best position of $\boldsymbol{x^{(i)}}$

End For;

$\boldsymbol{t \leftarrow 0}$ # Number of iterations (number of movements)

Repeat

  For $\boldsymbol{i \leftarrow 1}$ **to** $\boldsymbol{N}$ **do**

    Update the position of solution $\boldsymbol{x^{(i)}}$ as follows:

    IF $\left(\boldsymbol{x^* \neq x^{(i)}}\right)$ **THEN**

      For $\boldsymbol{j \leftarrow 1}$ **to** $\boldsymbol{D}$ **do**

        $\boldsymbol{v_j^{(i)} \leftarrow w * v_j^{(i)} + c_1 * \mathrm{rand}(0,1) * \left(x_j^{(*)} - x_j^{(i)}\right) + c_2 * \mathrm{rand}(0,1) * \left(x_j^{(i,p)} - x_j^{(i)}\right)}$ $\qquad \boldsymbol{v_j^{(i)} \in [-v_{max}, v_{max}],\ v_{max} = k(x_{max} - x_{min}),\quad k = 0.2}$

        $\boldsymbol{x_j^{(i)} \leftarrow x_j^{(i)} + v_j^{(i)}}$ $\qquad \boldsymbol{x_j^{(i)} \in [x_{min}, x_{max}]}$

      End For;

    End IF;

  End For;

  Evaluate the population $\boldsymbol{P}$ and update the best and personal solutions as follows:

  For $\boldsymbol{i \leftarrow 1}$ **to** $\boldsymbol{N}$ **do**

    IF $\boldsymbol{f(x^*) > f\left(x^{(i)}\right)}$ **THEN** $\boldsymbol{x^* \leftarrow x^{(i)}}$ **END IF;**

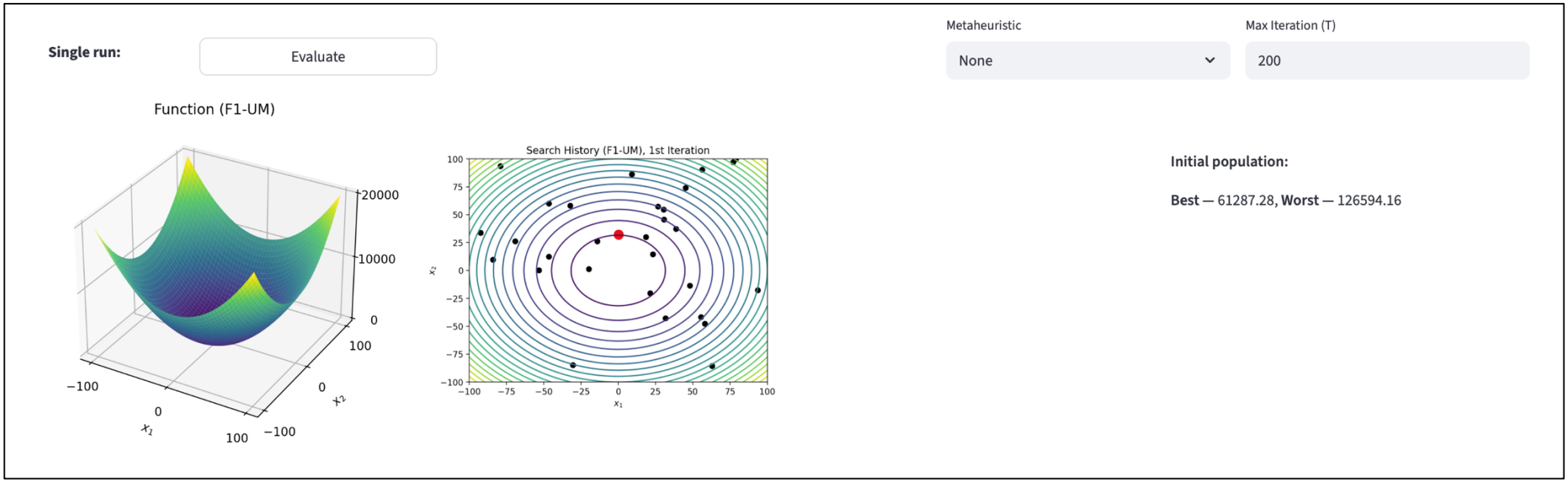
    IF $\boldsymbol{f\left(x^{(i,p)}\right) > f\left(x^{(i)}\right)}$ **THEN** $\boldsymbol{x^{(i,p)} \leftarrow x^{(i)}}$ **END IF;**

  End For;

  $\boldsymbol{t \leftarrow t + 1}$

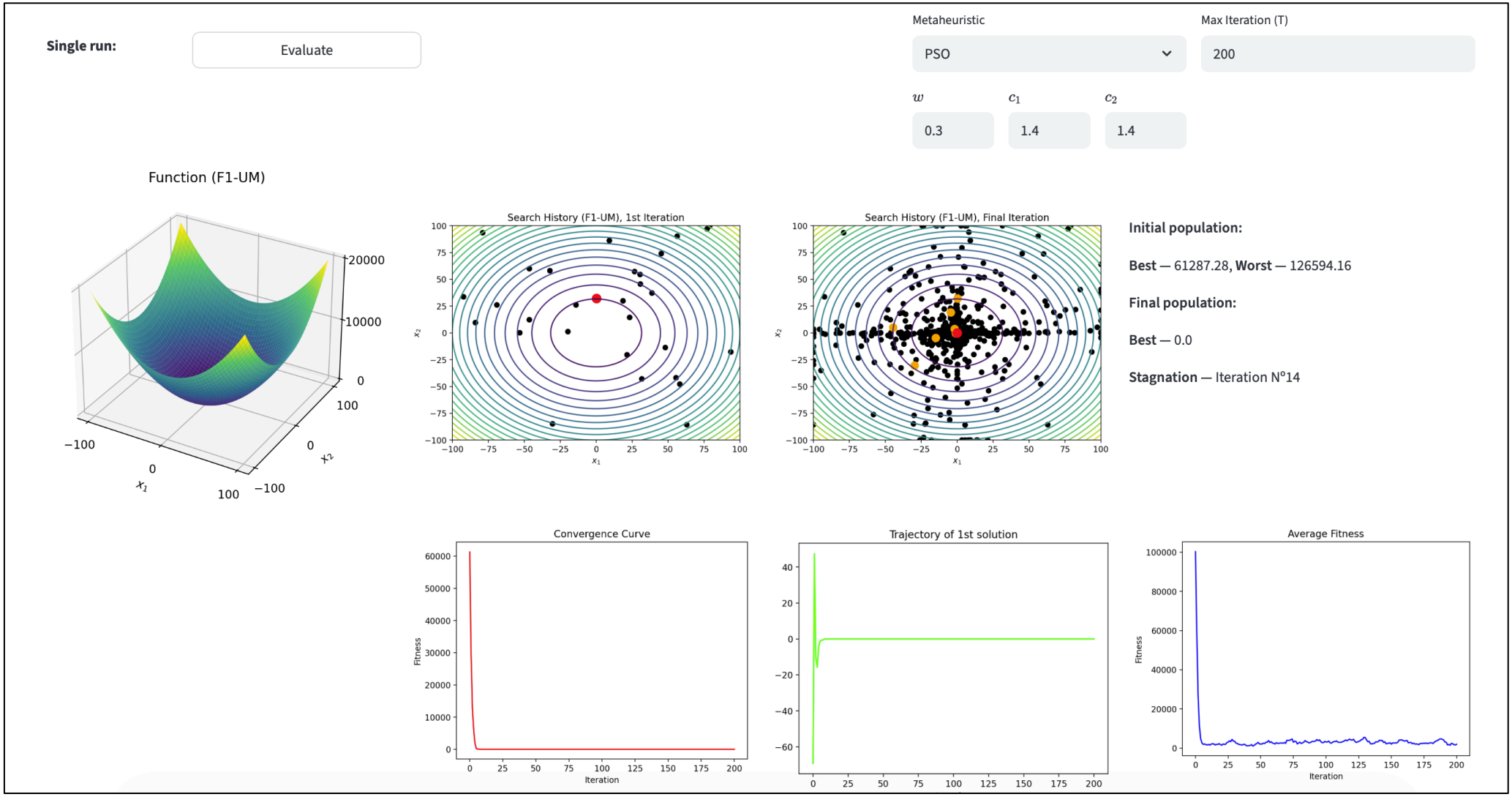
Until ($\boldsymbol{t = T}$) **or** (~~$\boldsymbol{x^*}$ did not change during the last three iterations~~)

Initial Population evaluation:

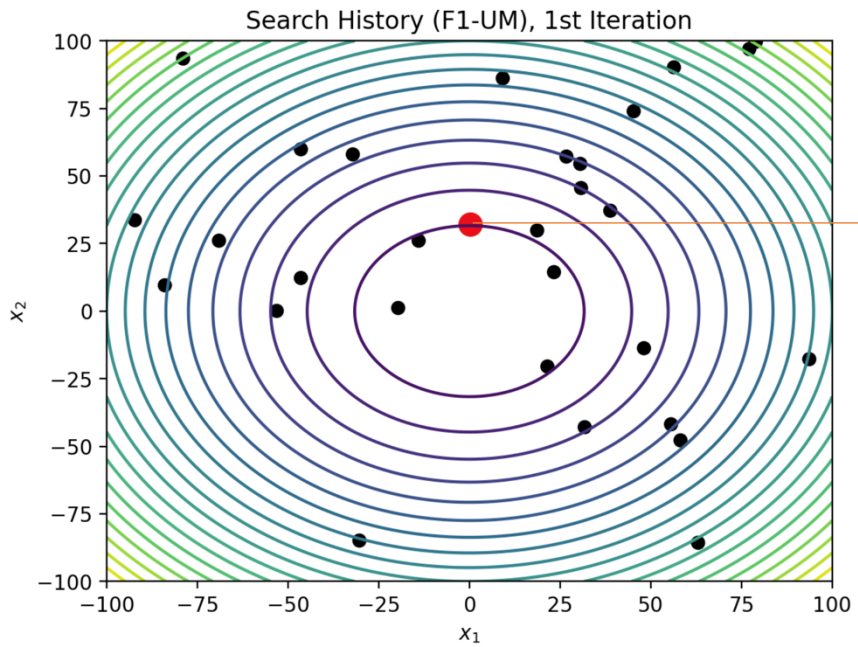


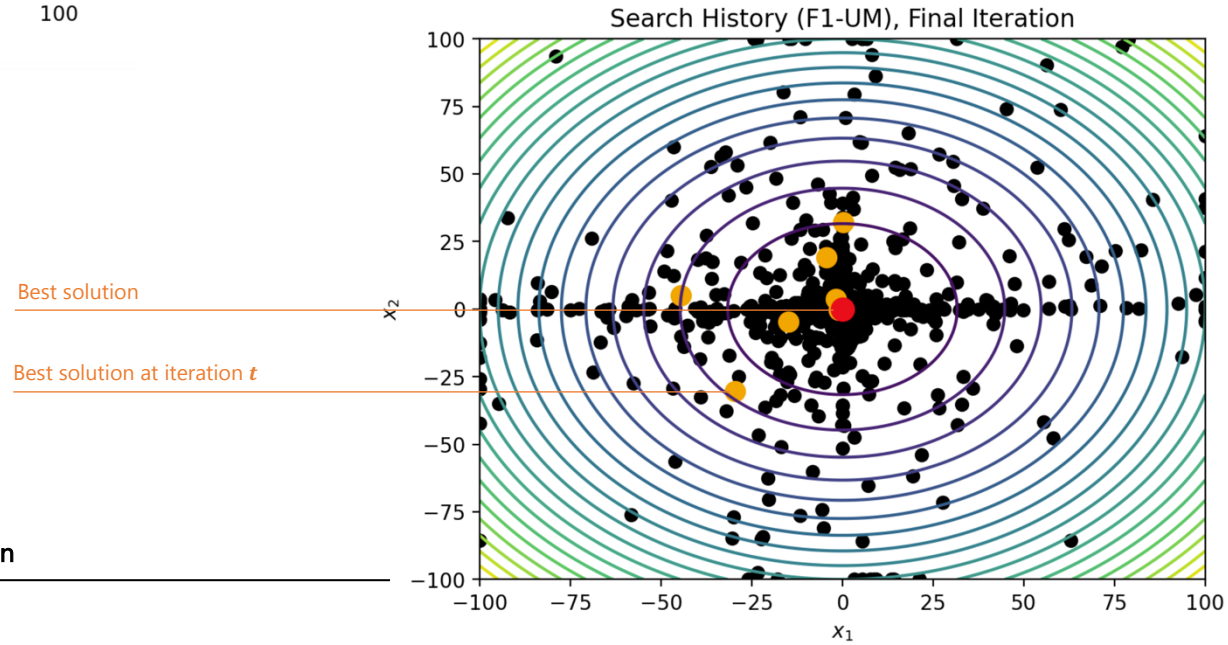
For more details, see TP N°2

Application of PSO for F1 function:

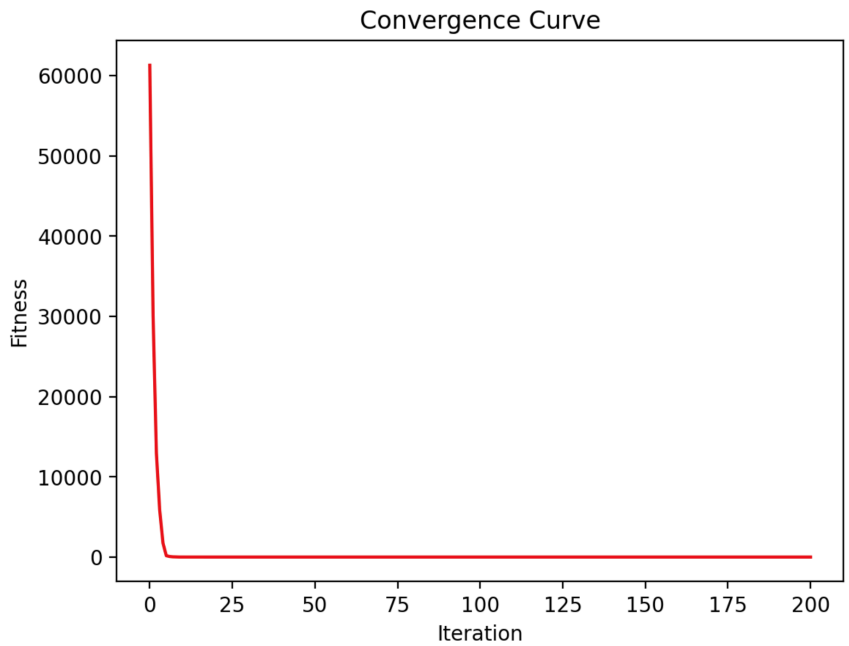


Initial population visualization



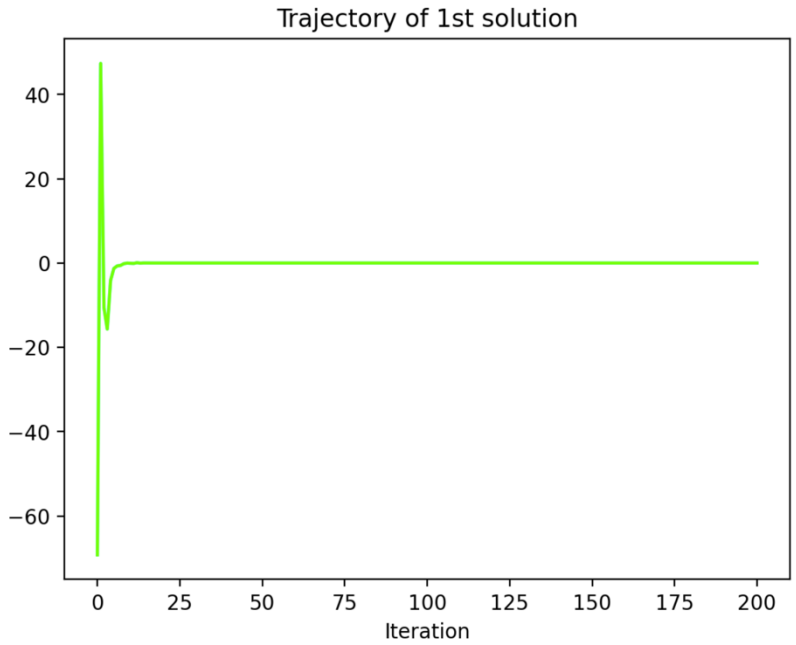


All position updates of the solutions in the initial population



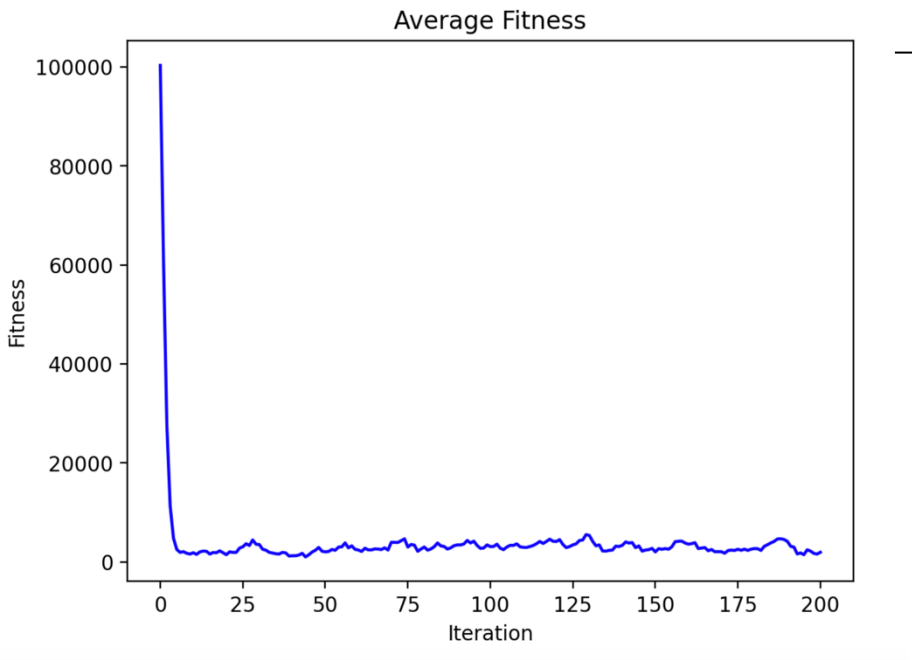
Convergence curve

Best fitness vs . Iteration



Trajectory of the first solution in the population

$x_1^{(1)}$ vs. Iteration



Average fitness of population

Average fitness of population vs . Iteration

Initial population:

Best — 61287.28, **Worst** — 126594.16

Final population:

Best — 0.0

Stagnation — Iteration N°14

Results

Stagnation corresponds to the iteration where the last improvement in fitness was observed.